



Bases

Theorem 1. *Every vector space V over a field k has a basis.*

Proof. Order the linearly independent subsets of V by inclusion. The union of a chain of such subsets is linearly independent, so Zorn's lemma gives a maximal linearly independent subset B . If $v \in V$ were outside the span of B , then $B \cup \{v\}$ would be linearly independent, against the maximality of B . Hence B spans V . $\square$



Dimension

Theorem 2. *Any two bases of V have the same cardinality.*

Proof. For finite bases, exchange the vectors of one basis for the vectors of the other, one at a time: each exchange keeps a spanning set, so neither basis has more vectors than the other. $\square$